

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

Practical – I

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Class:TYIT	Batch:1
Date of Assignment: 20/06/2026	Date/Time of Submission: 20/06/2026

AIM: Environment Setup and JavaScript Basics

A. Install Node.js and Visual Studio Code, and verify installation.

OUTPUT:

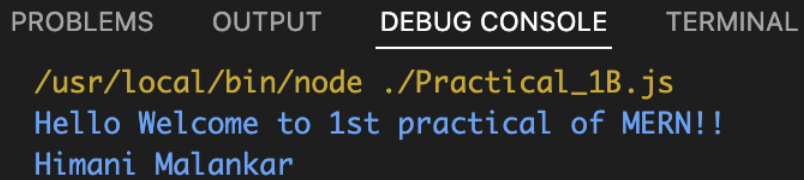
```
● himanimalankar@Himanis-Macbook MERN Practical % node -v
v24.17.0
● himanimalankar@Himanis-Macbook MERN Practical % npm -v
11.13.0
❗ himanimalankar@Himanis-Macbook MERN Practical %
```

B. Create a simple Node.js file and run it using the terminal.

CODE:

```
console.log("Hello Welcome to 1st practical of MERN!!!")
console.log("Himani Malankar")
```

OUTPUT:



The screenshot shows a web IDE interface with four tabs: PROBLEMS, OUTPUT, DEBUG CONSOLE, and TERMINAL. The DEBUG CONSOLE tab is active and displays the output of the Node.js script. The first line shows the command `/usr/local/bin/node ./Practical_1B.js` in yellow. The subsequent two lines show the output of the `console.log` statements in blue: `Hello Welcome to 1st practical of MERN!!` and `Himani Malankar`.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL
/usr/local/bin/node ./Practical_1B.js
Hello Welcome to 1st practical of MERN!!
Himani Malankar
```

C. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

CODE:

```
console.log("Basic Calculator...")

//Assigning Values
let a = 11;
let b = 11;

console.log("Addition: ", a+b)
console.log("Substraction: ", a-b)
console.log("Multiplication: ",a*b)
console.log("Divison: ",a/b)
console.log("Modulas: ",a%b)
```

OUTPUT:

```
[Running] node "/Users/himanimalkar/Desktop/MERN Practical/Practical_1C.js"
Basic Calculator...
Addition:  22
Substraction:  0
Multiplication:  121
Divison:  1
Modulas:  0
```

D. Create and run a simple program using variables and functions.

CODE:

```
let num1 = 15;
let num2 = 25;

function add(a , b){
  return a + b;
}

function multiply(a , b){
  return a * b;
}

console.log("1st number: ", num1);
console.log("2nd number: ",num2);

console.log("Addition = ", add(num1 , num2));
console.log("Multiplication = ", multiply(num1 , num2))
```

OUTPUT:

```
[Running] node "/Users/himanimalankar/Desktop/MERN Practical/Practical_1D.js"
1st number: 15
2nd number: 25
Addition = 40
Multiplication = 375
```

E. Demonstrate use of operators and control statements in JavaScript

CODE:

```
let a = 30;
let b = 20;
let day = 2;

//Comparison Operators
console.log("\nComparison Operators");
console.log("a > b :", a > b);
console.log("a < b :", a < b);
console.log("a == b :", a == b);
console.log("a != b :", a != b);

//if-else Statement
console.log("\nIf-Else Statement");
if (a > b){
    console.log("a is greater than b");
}else{
    console.log("b is greater than a or equal to a");
}

console.log("\nSwitch Statement");
switch(day){
    case 1:
        console.log("Monday");
        break;

    case 2:
        console.log("Tuesday");
        break;

    case 3:
        console.log("Wednesday");
        break;

    default:
        console.log("Invalid Day");
}
```

OUTPUT:

```
[Running] node "/Users/himanimalankar/Desktop/MERN Practical/Practical_1E.js"

Comparison Operators
a > b : true
a < b : false
a == b : false
a != b : true

If-Else Statement
a is greater than b

Switch Statement
Tuesday
```

F. Write programs using loops and template literals.

CODE:

```
let num = 20;
console.log(`Multiplication Table of ${num}`);

for (let i = 1 ; i <=10 ; i++){
  console.log(`${num} X ${i} = ${num * i}`);
}
```

OUTPUT:

```
[Running] node "/Users/himanimalankar/Desktop/MERN Practical/Practical_1F.js"
Multiplication Table of 20
20 X 1 = 20
20 X 2 = 40
20 X 3 = 60
20 X 4 = 80
20 X 5 = 100
20 X 6 = 120
20 X 7 = 140
20 X 8 = 160
20 X 9 = 180
20 X 10 = 200

[Done] exited with code=0 in 0.217 seconds
```